

ETF3231/5231

Business forecasting

Week 7: ARIMA models
<https://bf.numbat.space/>



Outline

- 1 Stationarity and differencing
- 2 Backshift notation

ARIMA models

- AR:** autoregressive (lagged observations as inputs)
- I:** integrated (differencing to make series stationary)
- MA:** moving average (lagged errors as inputs)

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An ARIMA model is rarely interpretable in terms of visible data structures like trend and seasonality. But it can capture a huge range of time series patterns.

Make data stationary (variance & mean), fit model, reverse, forecast.

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Definition

If $\{y_t\}$ is a **stationary time series**, then for all s , the distribution of $(y_t, \dots, y_{t+s})$ does not depend on t .

A **stationary series** is:

- roughly horizontal
- constant variance
- no patterns predictable in the long-term

Definition

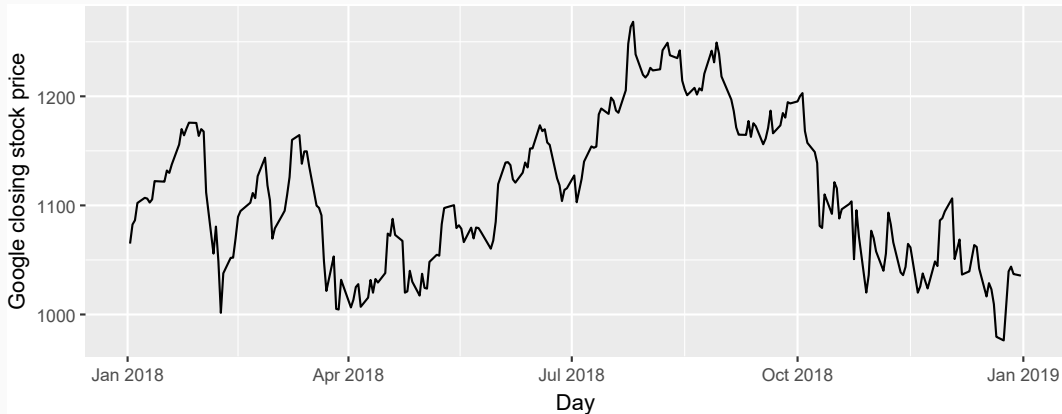
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- no patterns predictable in the long-term
- Transformations help to **stabilize the variance**.
- For ARIMA modelling, we also need to **stabilize the mean**.

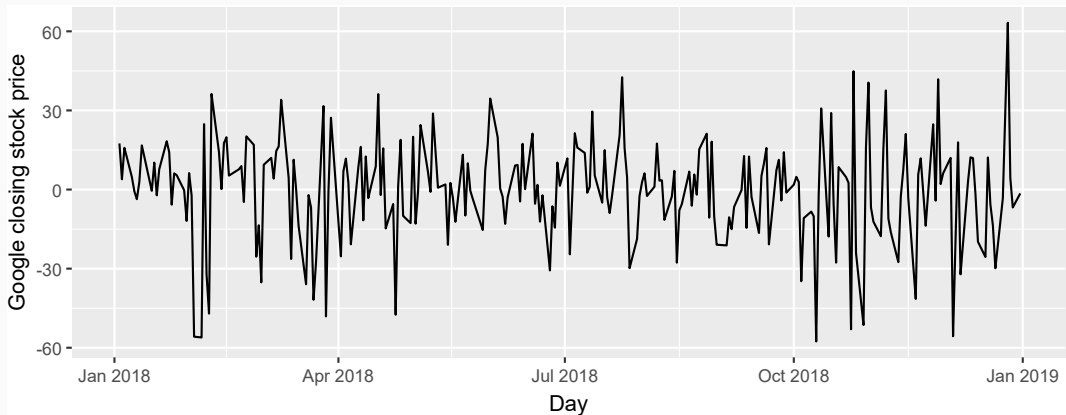
Stationary?

```
gafa_stock |>  
  filter(Symbol == "GOOG", year(Date) == 2018) |>  
  autoplot(Close) +  
  labs(y = "Google closing stock price", x = "Day")
```



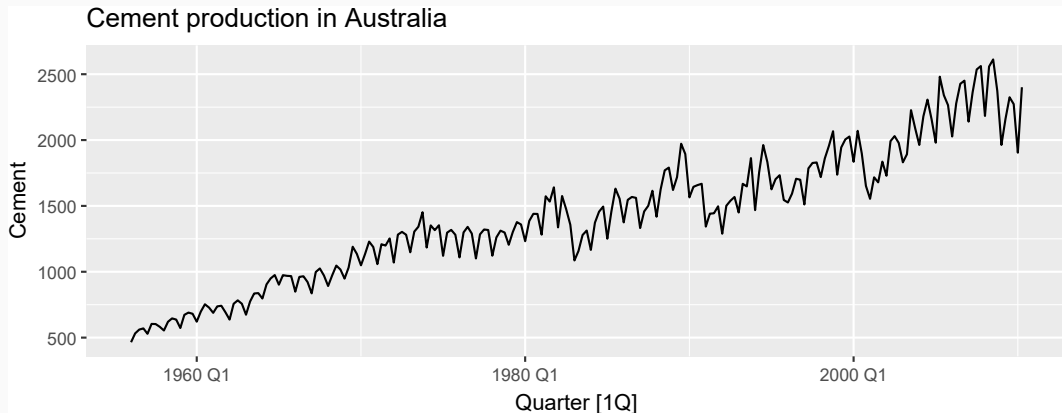
Stationary?

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gafa_stock |>  
  filter(Symbol == "GOOG", year(Date) == 2018) |>  
  autoplot(difference(Close)) +  
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```



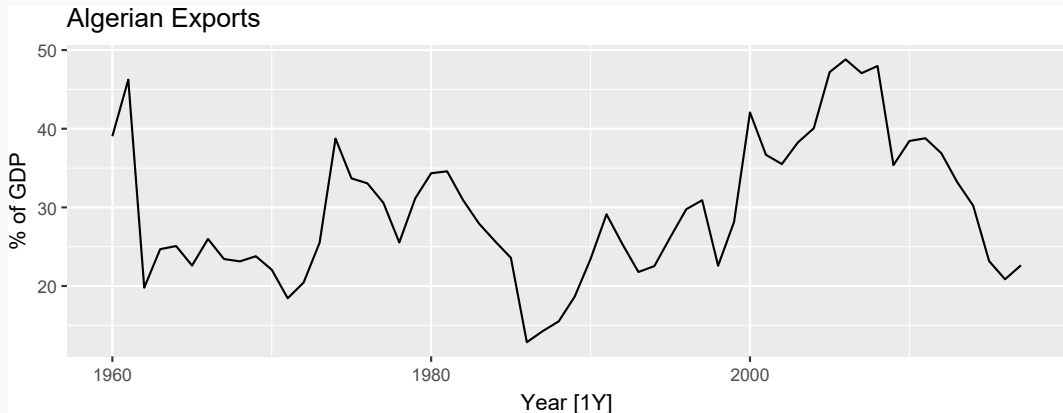
Stationary?

```
aus_production |>  
  autoplot(Cement) +  
  labs(title = "Cement production in Australia")
```



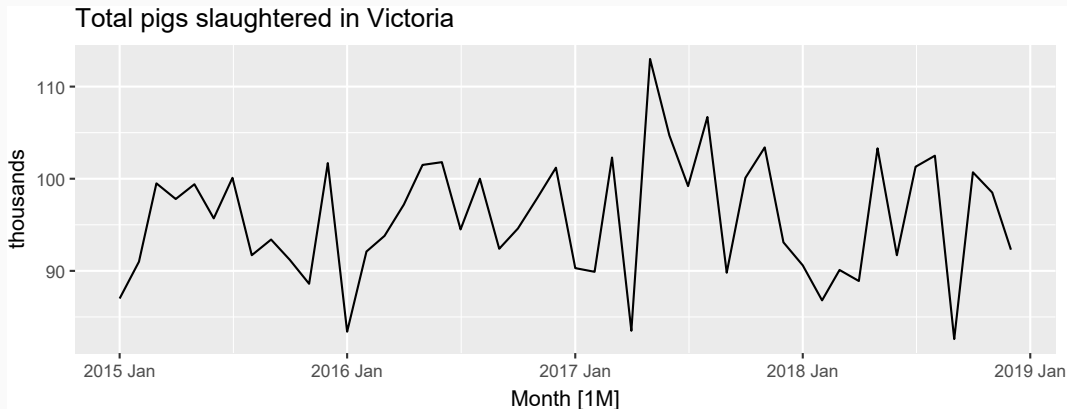
Stationary?

```
global_economy |>  
  filter(Country == "Algeria") |>  
  autoplot(Exports) +  
  labs(y = "% of GDP", title = "Algerian Exports")
```



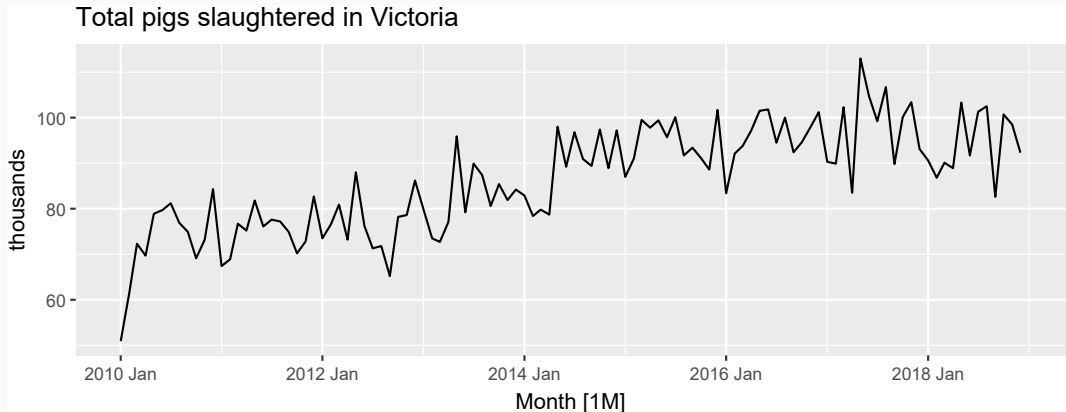
Stationary?

```
aus_livestock |>
  filter(Animal == "Pigs", State == "Victoria", year(Month) >= 2015) |>
  autoplot(Count/1e3) +
  labs(y = "thousands", title = "Total pigs slaughtered in Victoria")
```



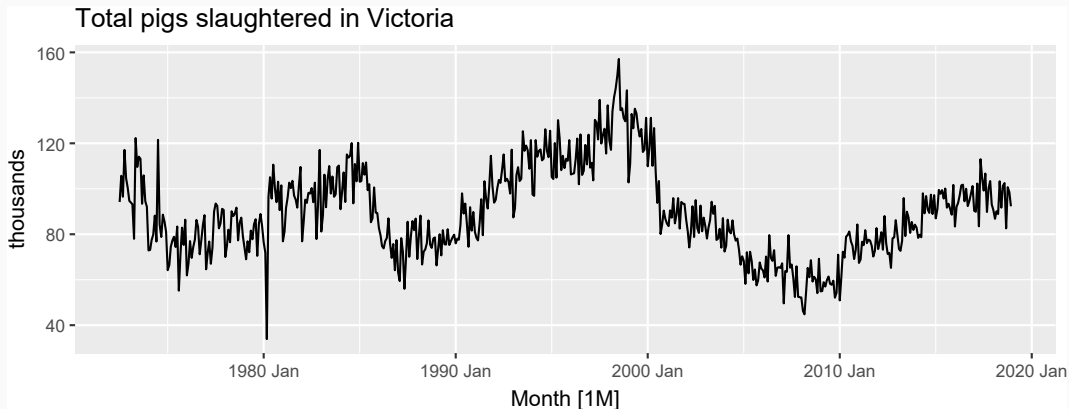
Stationary?

```
aus_livestock |>  
  filter(Animal == "Pigs", State == "Victoria", year(Month) >= 2010) |>  
  autoplot(Count/1e3) +  
  labs(y = "thousands", title = "Total pigs slaughtered in Victoria")
```



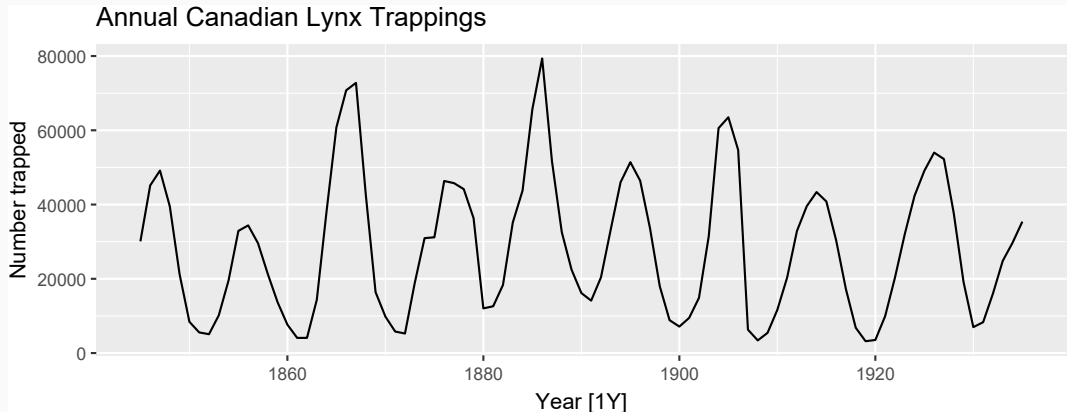
Stationary?

```
aus_livestock |>  
  filter(Animal == "Pigs", State == "Victoria") |>  
  autoplot(Count/1e3) +  
  labs(y = "thousands", title = "Total pigs slaughtered in Victoria")
```



Stationary?

```
pelt |>
  autoplot(Lynx) +
  labs(y = "Number trapped",
       title = "Annual Canadian Lynx Trappings")
```

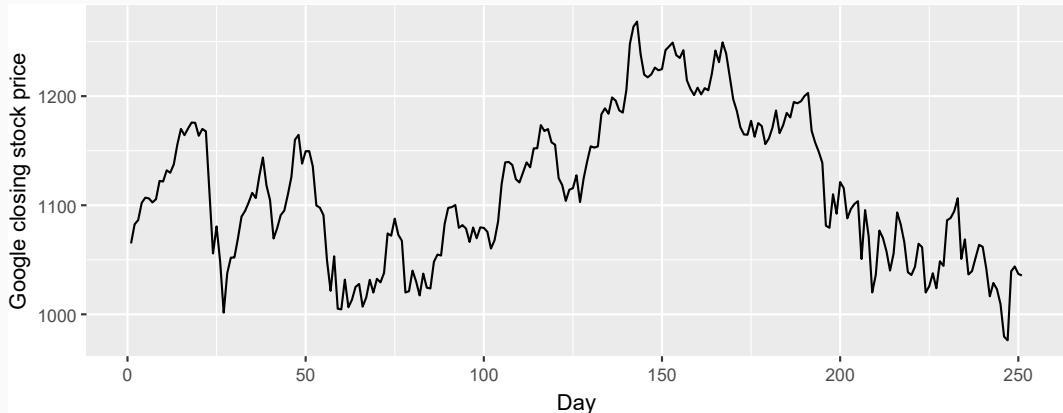


Example: Google stock price

```
google_2018 <- gafa_stock |>  
  filter(Symbol == "GOOG", year(Date) == 2018) |>  
  mutate(trading_day = row_number()) |>  
  update_tsibble(index = trading_day, regular = TRUE)
```

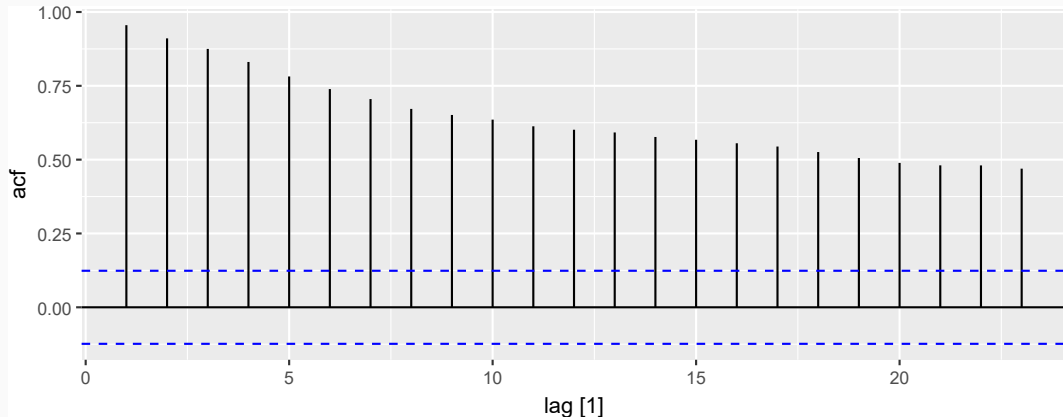
Example: Google stock price

```
google_2018 |>  
  autoplot(Close) + labs(y = "Google closing stock price", x = "Day")
```



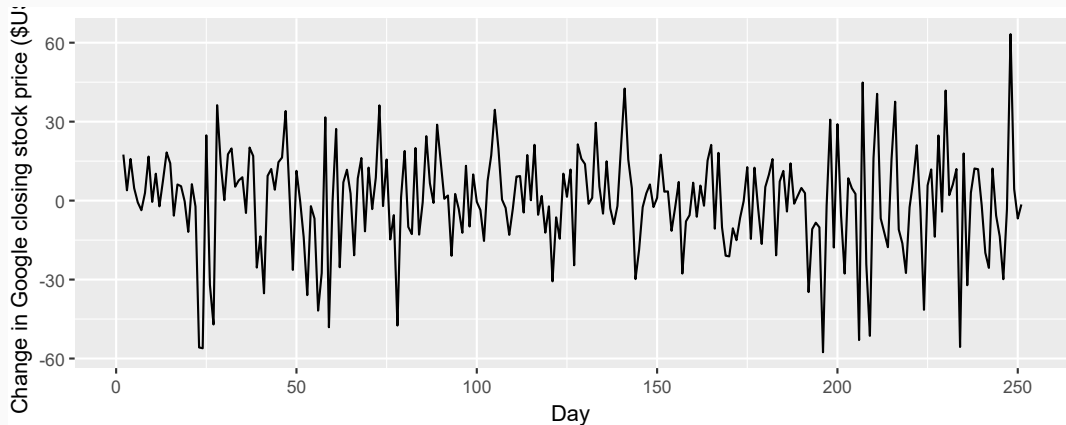
Example: Google stock price

```
google_2018 |>  
  ACF(Close) |> autoplot()
```



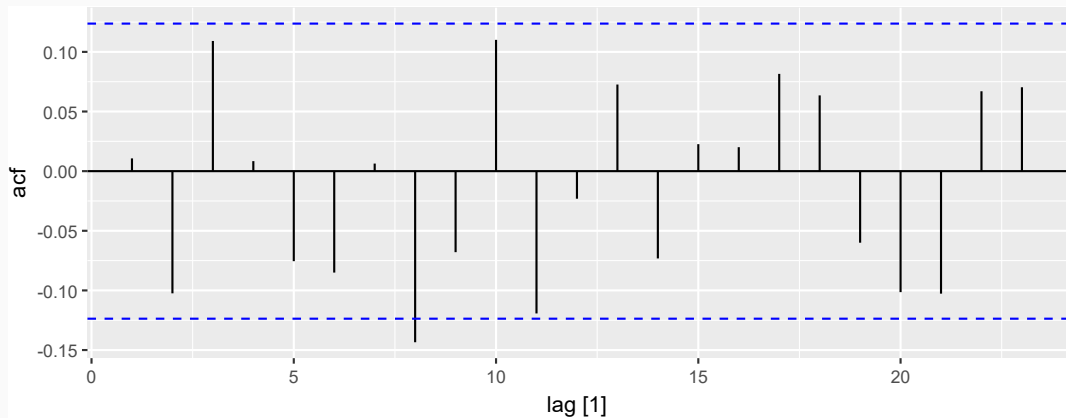
Example: Google stock price

```
google_2018 |>  
  autoplot(difference(Close)) +  
  labs(y ="Change in Google closing stock price ($USD)", x="Day")
```



Example: Google stock price

```
google_2018 |> ACF(difference(Close)) |> autoplot()
```



Differencing

- Differencing helps to **stabilize the mean**.
- The differenced series is the *change* between each observation in the original series: $y'_t = y_t - y_{t-1}$.
- The differenced series will have **only $T - 1$ values** since it is not possible to calculate a difference y'_1 for the first observation.

Example: Google stock price

- The differences are the **day-to-day** changes.
- Now the series looks just like a white noise series:
 - ▶ No autocorrelations outside the 95% limits.
 - ▶ Large Ljung-Box p-value.
- **Conclusion:** The daily change in the Google stock price is essentially a random amount uncorrelated with previous days.

Random walk model

- Graph of differenced data suggests the following model:

$$y_t - y_{t-1} = \varepsilon_t \quad \text{or} \quad y_t = y_{t-1} + \varepsilon_t$$

where $\varepsilon_t \sim NID(0, \sigma^2)$.

- Very widely used for non-stationary data.
- This is **the model behind the naïve method**.
- Random walks typically have:
 - ▶ long periods of apparent trends up or down.
 - ▶ Sudden/unpredictable changes in direction - **stochastic trend**.
- Forecast are equal to the last observation (Naïve)
 - ▶ future movements are unpredictable - movements up or down are equally likely.

Random walk with drift model

- If the differenced series has a non-zero mean then:

$$y_t - y_{t-1} = c + \varepsilon_t \quad \text{or} \quad y_t = c + y_{t-1} + \varepsilon_t$$

where $\varepsilon_t \sim NID(0, \sigma^2)$.

- c is the **non-zero average change** between consecutive observations.
- If $c > 0$, y_t will tend to drift upwards and vice versa.
 - ▶ **Stochastic and deterministic trend.**
- This is **the model behind the drift method**.

Further differencing

- Occasionally you need to difference non-seasonal data twice.

Further differencing

- Occasionally you need to difference non-seasonal data twice.
- We **seasonally difference** seasonal data.

$$y'_t = y_t - y_{t-m}$$

where m = number of seasons.

- ▶ For monthly data $m = 12$, for quarterly data $m = 4$.
- ▶ Seasonally differenced series will have $T - m$ obs.

Seasonal random walk

If seasonally differenced data is white noise it implies:

$$y_t - y_{t-m} = \varepsilon_t \quad \text{or} \quad y_t = y_{t-m} + \varepsilon_t$$

- The model behind the seasonal naïve method.

Seasonal differencing

Common to take **both seasonal and first differences**. When both are applied...

Seasonal differencing

Common to take **both seasonal and first differences**. When both are applied...

- it makes no difference which is done first—the result will be the same.
- If seasonality is strong, we recommend that **seasonal differencing be done first** because sometimes the resulting series will be stationary and there will be no need for further first difference.

Unit root tests

Statistical tests to determine the required order of differencing.

- 1 Augmented Dickey Fuller test: null hypothesis is that the data are **non-stationary** and non-seasonal.
- 2 Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: null hypothesis is that the data are **stationary** and non-seasonal.
- 3 Other tests available for seasonal data.

Unit root tests

Statistical tests to determine the required order of differencing.

- 1 Augmented Dickey Fuller test: null hypothesis is that the data are **non-stationary** and non-seasonal. H_0 : non-stationary
- 2 Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test: null hypothesis is that the data are **stationary** and non-seasonal. H_0 : stationary
- 3 Other tests available for seasonal data.

Seasonal differencing

STL decomposition: $y_t = T_t + S_t + R_t$

Seasonal strength $F_s = \max\left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(S_t + R_t)}\right)$

If $F_s > 0.64$, do one seasonal difference.

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Backshift notation

- First-order difference is denoted as $(1 - B)y_t$;
- Second-order difference is denoted as $(1 - B)^2y_t$;
- Second-order difference is not the same as a second difference, which would be denoted $(1 - B^2)y_t$;
- In general, a d th-order difference can be written as $(1 - B)^d y_t$

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- Second-order difference is not the same as a second difference, which would be denoted $(1 - B^2)y_t$;
- In general, a d th-order difference can be written as $(1 - B)^d y_t$
- A seasonal difference is denoted as $(1 - B^m)y_t$;
- A seasonal difference followed by a first difference can be written as

$$(1 - B^m)(1 - B)y_t$$